placed in front of each slit; it marks any passing photon so that when the photon is examined later, you can tell through which slit it passed. The question of how you can place a mark on a photon—how you can do the equivalent of placing an "L" on a photon that passes through the left slit and an "R" on a photon that passes through the right slit—is a good one, but the details are not particularly important. Roughly, the process relies on using a device that allows a photon to pass freely through a slit but forces its spin axis to point in a particular direction. If the devices in front of the left and right slits manipulate the photon spins in specific but distinct ways, then a more refined detector screen that not only registers a dot at the photon's impact location, but also keeps a record of the photon's spin orientation, will reveal through which slit a given photon passed on its way to the detector.

When this double-slit-with-tagging experiment is run, the photons do not build up an interference pattern, as in Figure 7.4a. By now the explanation should be familiar: the new tagging devices allow which-path information to be gleaned, and which-path information singles out one history or another; the data show that any given photon passed through either the left slit or the right slit. And without the combination of left-slit and right-slit trajectories, there are no overlapping probability waves, so no interference pattern is generated.

Now, here is Scully and Drühl's idea. What if, just before the photon hits the detection screen, you eliminate the possibility of determining through which slit it passed by erasing the mark imprinted by the tagging device? Without the means, even in principle, to extract the which-path information from the detected photon, will both classes of histories come back into play, causing the interference pattern to reemerge? Notice that this kind of "undoing" the past would fall much further into the shocking category than the ballplayer's diving catch in the ninth inning. When the tagging devices are turned on, we imagine that the photon obediently acts as a particle, passing through the left slit *or* the right slit. If somehow, just before it hits the screen, we erase the which-slit mark it is carrying, it seems too late to allow an interference pattern to form. For interference, we need the photon to act like a wave. It must pass through both slits so that it can cross-mingle with itself on the way to the detector screen. But our initial tagging of the photon seems to ensure that it acts